

Problem 1: Print a Right-Angle Triangle of Stars

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a right-angle triangle pattern of stars (*). Each row should contain an increasing number of stars, starting from 1 star in the first row.

Example 1: Input: n = 4

Output:

```
*  
  
**  
  
***  
  
****
```

Problem 2: Print a Square of Stars

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a square pattern of stars (*). Each row and column should have the same number of stars.

Example 1: Input: n = 5

Output:

```
*****  
  
*****  
  
*****  
  
*****  
  
*****
```

Problem 3: Print a Pyramid Pattern

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a pyramid pattern with stars (*). The pyramid should have a single peak and each row should have an increasing number of stars, centered horizontally.

Example 1: Input: n = 3

Output:

```
*  
  
**  
  
***
```

Problem 4: Print a Diamond Pattern

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a diamond pattern with stars (*). The pattern should include a single peak in the middle with symmetric rows above and below it.

Example 1: Input: n = 3

Output:

```
*  
  
***  
  
*****  
  
***  
  
*
```

Problem 5: Print a Hollow Square of Stars

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a hollow square pattern with stars (*). The border of the square should be filled with stars while the inner part should be empty.

Example 1: Input: n = 5

Output:

```
*****  
  
*  *  
  
*  *  
  
*  *  
  
*****
```

Problem 6: Print a Number Triangle

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a right-angle triangle pattern with numbers. Each row should contain an increasing sequence of numbers starting from 1.

Example 1: Input: n = 4

Output:

```
1  
  
12  
  
123
```

Problem 7: Print an Inverted Triangle Pattern**Difficulty:** Easy**Topics:** Pattern Printing**Hint:** Print an inverted triangle pattern with stars (*). Each row should contain decreasing numbers of stars from the top row.**Example 1: Input:** n = 5**Output:**

```
*****
****
***
**
*
```

Problem 8: Print a Diamond Pattern with Numbers**Difficulty:** Easy**Topics:** Pattern Printing**Hint:** Print a diamond pattern with numbers. The pattern should have a peak in the middle with symmetric rows above and below it.**Example 1: Input:** n = 3**Output:**

```
1
121
12321
121
1
```

Problem 9: Print a Right-Angle Triangle of Numbers**Difficulty:** Easy**Topics:** Pattern Printing**Hint:** Print a right-angle triangle pattern with increasing numbers. Each row should contain a continuous sequence of increasing numbers.**Example 1: Input:** n = 4**Output:**

1

23

456

78910

Problem 10: Print a Pyramid Pattern with Numbers

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a pyramid pattern with increasing numbers. Each row should have an increasing sequence of numbers, centered horizontally.

Example 1: Input: n = 3

Output:

1

232

34543

Problem 11: Print a Pattern of Alternating 0s and 1s

Difficulty: Medium

Topics: Matrix Pattern

Hint: Print a matrix where elements alternate between 0 and 1. The pattern should alternate both row-wise and column-wise.

Example 1: Input: n = 4

Output:

0101

1010

0101

1010

Problem 12: Print a Pascal's Triangle

Difficulty: Medium

Topics: Matrix Pattern

Hint: Print Pascal's Triangle up to N rows. Each row should be constructed based on the sum of the elements directly above it in the previous row.

Example 1: Input: n = 4

Output:

```
1
1 1
1 2 1
1 3 3 1
```

Problem 13: Print a Pattern of Consecutive Numbers

Difficulty: Medium

Topics: Matrix Pattern

Hint: Print a matrix of consecutive numbers starting from 1, filling rows sequentially.

Example 1: Input: n = 3

Output:

```
1 2 3
4 5 6
7 8 9
```

Problem 14: Print a Star Pattern with Increasing Width

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern where each row has an increasing width of stars.

Example 1: Input: n = 3

Output:

```
*
***
*****
```

Problem 15: Print a Right-Angle Triangle Pattern with Characters

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a right-angle triangle pattern using characters. Each row should contain the same character repeated according to the row number.

Example 1: Input: n = 3

Output:

```
A
BB
```

CCC

Problem 16: Print a Checkerboard Pattern

Difficulty: Medium

Topics: Matrix Pattern

Hint: Print a checkerboard pattern with two different characters alternating.

Example 1: Input: n = 4

Output:

XOXOXO

OXOXOX

XOXOXO

OXOXOX

Problem 17: Print a Pyramid Pattern of Increasing Stars

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pyramid pattern where each row increases in the number of stars.

Example 1: Input: n = 3

Output:

*

Problem 18: Print a Border Pattern with Numbers

Difficulty: Medium

Topics: Matrix Pattern

Hint: Print a border pattern using numbers. The border should be filled with numbers, and the inner part should be empty.

Example 1: Input: n = 4

Output:

1234

1 1

1 1

1234

Problem 19: Print an Inverted Pyramid Pattern with Characters

Difficulty: Medium

Topics: Pattern Printing

Hint: Print an inverted pyramid pattern using characters. Each row should have decreasing characters from the top row.

Example 1: Input: n = 3

Output:

CCC

BB

A

Problem 20: Print a Cross Pattern with Stars

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a cross pattern using stars. The cross should be centered within a matrix.

Example 1: Input: n = 5

Output:

*

*

*

Problem 21: Print a Spiral Matrix

Difficulty: Hard

Topics: Matrix Pattern

Hint: Print a matrix filled with numbers in a spiral pattern. The numbers should start from 1 and increment as you move around the spiral.

Example 1: Input: n = 3

Output:

1 2 3

8 9 4

7 6 5

Problem 22: Print a Diamond Pattern with Increasing Width

Difficulty: Hard

Topics: Pattern Printing

Hint: Print a diamond pattern where each line has increasing width of stars.

Example 1: Input: n = 3

Output:

```
*  
  
***  
  
*****  
  
***  
  
*
```

Problem 23: Print a Diamond Pattern with Numbers Increasing

Difficulty: Hard

Topics: Pattern Printing

Hint: Print a diamond pattern where numbers increase. Each row should show a symmetrical pattern with numbers increasing towards the center.

Example 1: Input: n = 3

Output:

```
1  
121  
12321  
121  
1
```

Problem 24: Print a Pattern of Increasing and Decreasing Stars

Difficulty: Hard

Topics: Pattern Printing

Hint: Print a pattern where stars increase to a midpoint and then decrease.

Example 1: Input: n = 3

Output:

```
*  
  
***  
  
*****
```

*

Problem 25: Print a Matrix with Zigzag Pattern

Difficulty: Hard

Topics: Matrix Pattern

Hint: Print a matrix with a zigzag pattern of numbers. The numbers should alternate direction row-wise.

Example 1: Input: n = 3

Output:

1 2 3 4

8 7 6 5

9 10 11 12

Problem 26: Print a Pattern of Alternating Characters in Rows

Difficulty: Hard

Topics: Pattern Printing

Hint: Print a pattern where rows alternate between two characters.

Example 1: Input: n = 4

Output:

ABAB

BABA

ABAB

BABA

Problem 27: Print a Number Pyramid Pattern with Characters

Difficulty: Hard

Topics: Pattern Printing

Hint: Print a pyramid pattern using increasing characters, where each row increases in width and character range.

Example 1: Input: n = 3

Output:

A

BCD

EFGHI

Problem 28: Print a Pattern with Diagonal Lines

Difficulty: Hard

Topics: Pattern Printing

Hint: Print a pattern with diagonal lines using characters. Each diagonal line should be aligned properly.

Example 1: Input: n = 4

Output:

```
A
B B
C  C
D   D
```

Problem 29: Print a Matrix with Diamond Pattern of Numbers

Difficulty: Hard

Topics: Matrix Pattern

Hint: Print a matrix where elements follow a diamond pattern with numbers.

Example 1: Input: n = 3

Output:

```
1
121
12321
121
1
```

Problem 30: Print a Cross Pattern of Stars with Diagonals

Difficulty: Hard

Topics: Pattern Printing

Hint: Print a cross pattern using stars with intersecting diagonals.

Example 1: Input: n = 5

Output:

```
* * * * *
* * * *
* * *
```

```
* *  
  
*  
  
* *  
  
* * *  
  
* * * *  
  
* * * * *
```

Problem 31: Print a Triangular Matrix with Numbers

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a triangular matrix where each row contains increasing numbers. Each subsequent row should start from the next number.

Example 1: Input: n = 3

Output:

```
1  
2 3  
4 5 6
```

Problem 32: Print a Star Pattern with Increasing and Decreasing Width

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern with stars that increase to a midpoint and then decrease. The stars should be centered horizontally.

Example 1: Input: n = 3

Output:

```
*  
  
* * *  
  
* * * * *  
  
* * *  
  
*
```

Problem 33: Print a Pattern of Nested Squares

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern with nested squares using stars. The outermost square should be filled with stars, and each subsequent square should be smaller and centered inside the previous one.

Example 1: Input: n = 5

Output:

* *

* *

* *

Problem 34: Print a Pattern with Increasing Characters in Columns

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern where each column contains increasing characters from A.

Example 1: Input: n = 3

Output:

A

B C

D E F

Problem 35: Print a Matrix with Spiral Diagonals

Difficulty: Hard

Topics: Matrix Pattern

Hint: Print a matrix with numbers arranged in diagonal spirals. The numbers should fill the matrix in a diagonal spiral fashion.

Example 1: Input: n = 3

Output:

1 2 3

4 5 6

7 8 9

Problem 36: Print a Checkerboard Pattern with Increasing Size

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a checkerboard pattern where the size of each square increases as you move along the matrix.

Example 1: Input: n = 3

Output:

XOX

OXO

XOX

Problem 37: Print a Cross Pattern with Increasing Size

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a cross pattern where the size of the cross increases with each row. The pattern should be centered.

Example 1: Input: n = 3

Output:

*

*

Problem 38: Print a Pattern of Alternating Triangles

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern with alternating triangles of stars. The triangles should alternate direction.

Example 1: Input: n = 3

Output:

*

*

Problem 39: Print a Matrix with Diamond Pattern of Numbers

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a matrix where numbers form a diamond pattern. The numbers should increase and decrease symmetrically around the center.

Example 1: Input: n = 3

Output:

```
1
121
12321
121
1
```

Problem 40: Print a Star Pattern with Increasing Width and Centered

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern where the width of stars increases, and the stars are centered horizontally.

Example 1: Input: n = 3

Output:

```
*
***
*****
```

Problem 41: Print a Pattern with Spiral and Zigzag

Difficulty: Hard

Topics: Matrix Pattern

Hint: Print a matrix with both spiral and zigzag patterns. The matrix should first fill in a spiral pattern and then in a zigzag fashion.

Example 1: Input: n = 3

Output:

```
1 2 3
6 5 4
7 8 9
```

Problem 42: Print a Pattern of Alternating Characters in Matrix

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a matrix where characters alternate in each cell to form a pattern.

Example 1: Input: n = 3

Output:

ABAB

BABA

ABAB

Problem 43: Print a Pattern with Nested Triangles

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern with nested triangles of stars. Each triangle should be centered and decrease in size.

Example 1: Input: n = 3

Output:

*

*

Problem 44: Print a Matrix with Increasing Rows and Columns

Difficulty: Easy

Topics: Matrix Pattern

Hint: Print a matrix where each row and column contains increasing numbers.

Example 1: Input: n = 3

Output:

1 2 3

4 5 6

7 8 9

Problem 45: Print a Pattern with Rows of Increasing Characters

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern where each row contains an increasing sequence of characters.

Example 1: Input: n = 3

Output:

A

BC

DEF

Problem 46: Print a Star Pattern with Diamond Shape and Numbers

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern with a diamond shape of stars, with numbers inside the diamond.

Example 1: Input: n = 3

Output:

```
1
121
12321
121
1
```

Problem 47: Print a Matrix with Cross Pattern of Numbers

Difficulty: Hard

Topics: Matrix Pattern

Hint: Print a matrix where the center forms a cross pattern with numbers.

Example 1: Input: n = 5

Output:

```
12321
01210
01210
01210
12321
```

Problem 48: Print a Pattern with Concentric Squares

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a pattern with concentric squares using stars. The outer square should be larger and each subsequent square should be centered inside.

Example 1: Input: n = 5

Output:

* *

* *

* *

Problem 49: Print a Pattern of Alternating Rows and Columns of Numbers

Difficulty: Easy

Topics: Pattern Printing

Hint: Print a pattern with alternating rows and columns of numbers, where each row and column increases sequentially.

Example 1: Input: n = 3

Output:

123

456

789

Problem 50: Print a Matrix with Zigzag Pattern of Stars

Difficulty: Medium

Topics: Pattern Printing

Hint: Print a matrix where stars form a zigzag pattern, alternating rows in their positioning.

Example 1: Input: n = 3

Output:

* * *

* *

* * *